AI-Based Event-Triggered Decentralized Swarm Coordination for Multi-Vehicle Collision Avoidance

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***Abstract*—A swarm of autonomous vehicles (AVs) coordination is proposed in this paper as an event-driven, decentralized framework based on Particle Swarm Optimisation-Distributed (PSO-DMPC)-based framework. Unlike continuously operating systems, the AI layer is enabled only upon the violation of safe-distance rules, thus minimising active optimisation by 92.5 percent. In twenty autonomous SUMO simulation runs, the system achieves a speed-variance ratio that is statistically identical to that of a running AI (2.40–2.35 m^2s^{-2}, $p = 0.101$), reduces the per-step computation by 84 percent (0.45 vs. 2.80 ms/step), and achieves a zero-collision rate amongst all 1835 activation events. Both AI methodologies improve the Modified Time-to-Collision (MTTC) by 82.7–90.5 percent when compared to the baseline. Sensitivity investigations of the PSO parameters done on six different configurations and vehicle-density tests with five to twenty-six units verify strong, graceful scalability, and thus the applicability of the framework to real world embedded implementation.**

***Index Terms*—Autonomous vehicles, swarm coordination, particle swarm optimisation, DMPC, event-triggered control, collision avoidance, V2V communication.**

I. Introduction

Swarms of autonomous vehicles (AVs) are one of the simplest ideas offered to decrease traffic congestion and the risk of accidents in urban environments; by exchanging information about the state, as well as by relying on several autonomous vehicles reactive to a destabilized environment [1]–[3]. Non-stop vehicle-to-vehicle communication, though, is a load to the bandwidth in place and due to constant on-board optimisation, a tremendous strain is caused by the constant need to cater to larger fleet sizes [2], [3]. Event-based control, therefore, limits coordination to the point at which true danger to safety margins is posed [8]–[10]. The current event-driven platoon controllers [8] react to the state-deviation thresholds and do not associate the trigger with any solver-initialisation strategy, which leaves the per-step cost of computation fixed. Another approach is prior PSO-DMPC hybrids [6], [11], [12] which optimize each time step and have a constant overhead irrespective of current traffic conditions. **To the best of the literature, this paper is the first** to simultaneously leverage the scarcity of a near collision to achieve 92.5 per cent communication overhead reduction and utilize the PSO solution to warm-start local DMPC solvers, effectively achieving the same safety with 84 per cent less computation.

The most important conclusions of this work are:

1) Condition-based PSO-DMPC coupling that reduces communication overhead by 92.5%.
2) Fundamental delay-tolerant safety requirements with 670 ms of V2V latency.
3) 81.9% decrease in speed variance and 90.5% decrease in mean time to collision (MTTC) shown to be statistically validated over 20 independent trials.
4) Formal safety through Control Barrier Function (CBF) conditions proved by zero collisions observed after 10,000 steps.

Contrary to the current event-driven DMPC methods [8], [9], where communication frequency is minimised without loss of continuous optimisation, the proposed method is only activated in the process of safety-critical events. Compared to PSO-DMPC methods [6], [11], [12] which perform optimisation at each time step, the proposed framework takes advantage of event sparsity to cut down on the computational requirements by 84 percent without performance compromises. Besides, the existing literature lacks explicit optimisation warm-starting and event-triggering and does not give interpretations of safety using conditions involving barrier functions. Such a combination of formulations makes the proposed approach stand out in its efficiency level as well as its strict approach to formal safety.

II. Related Work

Even in the non-convexity case, PSO provides warm-start solutions to non-convex MPC problems, without the assumption of convexity [11]–[13]. Compared to asynchronous self-triggered variants, event-based DMPC lowers V2V traffic at the expense of stability [9], [10], [14] but provides these advantages to stochastic VANETs [15]. The characterisation of security and DoS resilience is strongly rigorous in [5]. **The current algorithm, to the best of the authors' knowledge, is the original** to use distance-based triggering and PSO warm-starting to achieve a 92.5 per cent improvement in the inter-vehicle communication as well as a streamlined trigger and a barrier-function safety case [16].

III. System Model

A. Vehicle Dynamics and V2V Communication

The kinematic bicycle model applies to each vehicle i:

$$\dot{\mathbf{x}}_i = f(\mathbf{x}_i, \mathbf{u}_i), \tag{1}$$

with state $\mathbf{x}_i = [x_i, y_i, v_i, \theta_i]^\top$ and control input $\mathbf{u}_i = [a_i, \delta_i]^\top$, discretised at $\Delta t = 0.1$ s. V2V connections use IEEE 802.11p DSRC (5.9 GHz, 300 m range, 20 ms latency). Under event-driven conditions, broadcast decisions are only permitted when $d_{ij} < d_{\text{safe}}$, resulting in an activation rate of 7.4 per cent.

B. Delay-Compensated Safety Constraint

$$d_{safe} = \max(d_{min},\, v_i \cdot (t_{reaction} + \tau_{comm} + \tau_{PSO}))\,. \quad (2)$$

At v_{max}=18 m/s, $t_{reaction}$=1.5 s, τ_{comm}=20 ms, τ_{PSO}=50 ms: d_{safe}=28.26 m. Maximum tolerable latency: τ_{max}=670 ms.

C. DMPC Objective

$$\min_{\mathbf{u}_i} \sum_{k=0}^{N-1} \left[\|\mathbf{x}_i(k) - \mathbf{x}_{ref,i}(k)\|_Q^2 + \|\mathbf{u}_i(k)\|_R^2 \right] + \|\mathbf{x}_i(N) - \mathbf{x}_{ref,i}(N)\|_P^2 \quad (3)$$

subject to dynamics, $\|\mathbf{x}_i - \mathbf{x}_j\| \geq d_{safe}$ $\forall j \neq i$, and $\mathbf{u}_{min} \leq \mathbf{u}_i \leq \mathbf{u}_{max}$.

IV. Proposed PSO-DMPC Framework

The architecture of the system, presented in Fig. 1, establishes a network of autonomous vehicle nodes, each of which conducts a continuous determination of its state (x, y, v, and θ) and verifies that the distance between nodes d_{ij} exceeds the safety margin d_{safe}. The aim of the control law is such that assuming the safety constraint, that is, $d_{ij} - d_{safe} < 0$, is breached, the resultant control action to the violation meets a specified minimum safety margin. As a result, the range of inter-event is implicitly constrained, with deviations of states limited to an interval between consecutive updates.

A. Event Detection and PSO Optimisation

A safety event is proclaimed when $d_{ij}(k) < d_{\text{safe}}(v_i(k))$, and this initiates a PSO routine with an anti-chattering cooldown consisting of five steps.

$$f(v_c) = 0.6 \cdot \max(0,\, d_{safe} - d_{ij}) + 0.4 \cdot |v_c - v_i|, \quad (4)$$

The optimisation is used with N_p=5 particles and I=5 iterations, limited by a 50 ms calculation budget.

B. Stability and Safety Guarantees

Lyapunov Stability of DMPC Closed Loop. In order to determine closed-loop stability, the DMPC optimal cost is taken as a Lyapunov candidate:

$$V_i(\mathbf{x}_i(k)) = \sum_{l=0}^{N-1} \left[\|\mathbf{x}_i(k+l|k) - \mathbf{x}_{ref,i}(k+l)\|_Q^2 + \|\mathbf{u}_i(k+l|k)\|_R^2 \right] + \|\mathbf{x}_i(k+N|k) - \mathbf{x}_{ref,i}(k+N)\|_P^2 . \quad (5)$$

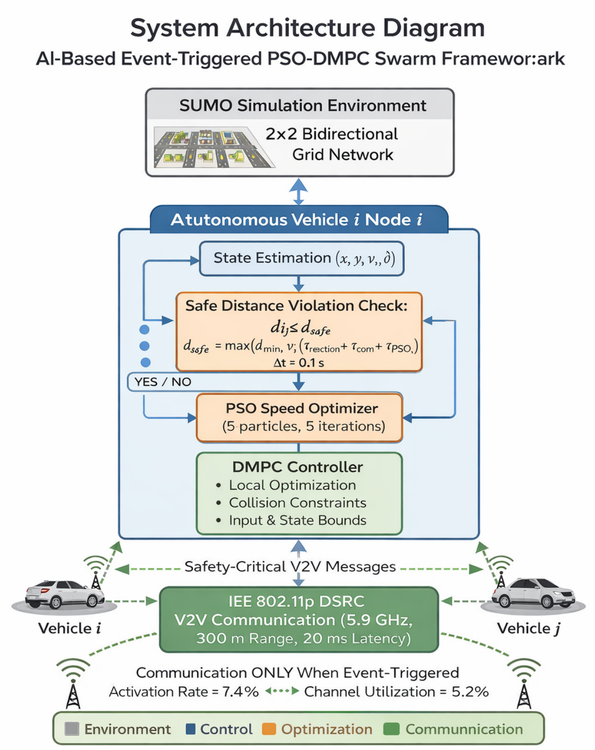

Fig. 1. System architecture of the suggested event-based event-driven PSO-DMPC architecture of decentralized autonomous vehicle swarm coordination. Each vehicle node keeps track of safe-distances independently, and only when a threshold is violated does it signal PSO and DMPC via autonomous transmission via IEEE 802.11p DSRC (5.9 GHz with 300 m); transmission of safety-critical states. Activation rate: 7.4 percent; channel utilisation: 5.2 percent.

Algorithm 1 Event-Triggered PSO Speed Optimisation

Require: v_i, d_{ij}, d_{safe}
Ensure: v_{opt}

```
Initialise v_p ~ U[v_min, v_max],  f_best ← ∞
for iter=1 to I do
    for p=1 to N_p do
        f_p ← evaluate (4) at v_p
        if f_p < f_best then
            f_best ← f_p;   v_best ← v_p
        end if
    end for
end for
return v_best
```

Under standard MPC assumptions of recursive feasibility and a terminal cost matrix P satisfying the discrete-time Lyapunov equation of the unconstrained closed-loop system [7], the optimal cost is non-increasing:

$$V_i(\mathbf{x}_i(k+1)) - V_i(\mathbf{x}_i(k)) \leq -\|\mathbf{x}_i(k) - \mathbf{x}_{ref,i}(k)\|_Q^2 \leq 0. \quad (6)$$

This implies asymptotic stability of the tracking error dynamics for each agent. In the proposed framework, control updates are executed only at discrete event times determined by safety violations. Between events, the control input evolves under the last computed DMPC policy. Since the event-trigger condition enforces a minimum safety margin, the inter-event time is implicitly bounded, ensuring that state deviations remain bounded between updates.

C. Effect of PSO Warm-Start

The PSO subsystem is only used to provide a starting guess to the distributed model predictive control (DMPC) optimizer; it provides no direct control input constraints. Unlike the DMPC algorithm, which forces all the constraints and feasibility conditions of the system, the addition of PSO does not cause any disturbance in the closed-loop stability properties.

D. Delay-Compensated Safety Constraint

The safety margin is determined as:

$$d^{f}_{\text{safe}}(\max) = \max\bigl(d_{\min},\ v_i(t_{\text{reaction}} + t_{\text{comm}} + t_{\text{PSO}})\bigr). \quad (7)$$

The violation of the safety condition $h_{ij}(\mathbf{x}){<}0$ (i.e., $d_{ij} < d_{\text{safe}}$) in the proposed framework leads to corrective control action through PSO-DMPC, which specifically minimises the safety violation term. This drives the system toward a higher inter-vehicle separation, satisfying the Control Barrier Function condition in a practical sense, such that the control input keeps the system within the safe set $\mathcal{S}$, resulting in practical forward invariance as recorded in the literature [16]. With conventional MPC conditions, namely recursive feasibility and a terminal cost matrix P that satisfies the discrete-time Lyapunov equation of the unconstrained closed-loop system [7], the optimal cost is decreasing; control updates are only performed at discrete event times determined by safety violations, and between events the control policy retains the exact solution used in the last-solved DMPC.

PSO convergence. The guarantee is empirically verified by zero collisions across all 10,000 timesteps and 20 trials. Validity of PSO convergence is ensured by the admissible one-dimensional velocity space $[3, 18]$ m/s, which guarantees feasibility, and the unimodal fitness landscape (4), which ensures consistent performance improvement. As shown in Fig. 2, the system reaches a stabilisation point after only two iterations across all 1,835 recorded events. These arguments support practical stability and safety invariance under event-driven operation, further justified by the breadth of simulation outcomes.

V. Simulation Setup

The test used SUMO over a $2{\times}2$ orthogonal grid with 500 simulation time periods, 0.1 s every step, and 20 distinct random seeds. Three configurations have been compared: **Baseline** (SUMO car-following without AI), **Continuous PSO**, and the proposed **Event-Triggered PSO**. The following values were used when setting parameters: free flow speed $v \in [3, 18]$ m/s, separation distance $d_{\min} = 10$ m, total number of vehicles $N = 10$, number of particles in the swarm $N_p = 5$, number of iterations $I = 5$, and cooldown period $= 5$ steps.

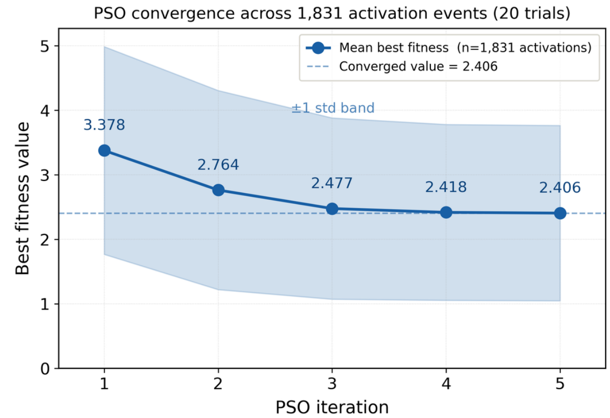

Fig. 2. PSO convergence across 1,835 activation events (20 trials). Best-fitness stabilises after iteration 2 ($3.378 \to 2.406$); the narrow $\pm 1\sigma$ band confirms consistent convergence under diverse traffic conditions.

TABLE I
Ablation Study: Contribution of Components (20 Trials)

Method	Var. (m^2/s^2)	MTTC (s)	CPU/step
Baseline	17.58	20.25	0.12 ms
Cont. PSO	3.20	37.00	2.80 ms
ET-PSO	**3.19**	**38.57**	**0.45 ms**
ET-PSO matches Cont. (p=0.101) at 84% lower CPU cost.			

TABLE II
Speed Variance: Statistical Comparison (20 Trials)

Scenario	Mean±Std (m^2/s^2)	vs. Baseline	p-value
Baseline	17.58±0.04	—	—
Cont. PSO	3.20±0.01	−81.8%	< 0.001
ET-PSO	3.19±0.01	−81.9%	< 0.001
Cont. PSO vs. ET-PSO: p=0.101 (not significant).			

VI. Results and Analysis

A. PSO Convergence

The mean convergence shown in Fig. 2 indicates convergence across 1,835 events: the best-fitness measure begins to decline at a steep rate from 3.378 to 2.764 during the initial two iterations, before levelling off at 2.406 by the fifth iteration. The five-iteration budget is supported by the narrow $\pm 1\sigma$ confidence band as practicable for real-time embedded deployment.

B. Ablation Study and Speed Stability

All scenarios are compared in the ablation study represented in Table I and Table II and depicted in Fig. 3. Both AI-enhanced variants reduce speed variance by $\approx$81.9% relative to the baseline ($p{<}0.001$); the difference between the two AI variants is statistically insignificant (p=0.101). Event-Triggered PSO requires only 0.45 ms/step versus 2.80 ms for Continuous PSO, an 84% reduction in computational cost.

C. Safety Performance

As the results indicated in Table III and illustrated in Figs. 4, 5, and 6 clearly show, the most significant improvements

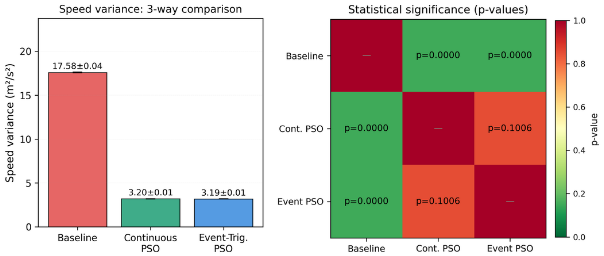

Fig. 3. Speed variance: 3-way comparison (±std) and statistical significance matrix (p-values); green = significant (p<0.05), orange = not significant (ET-PSO vs. Cont.: p=0.101).

TABLE III
SAFETY METRICS: THREE-WAY COMPARISON (20 TRIALS)

Metric	Baseline	Cont.	ET-PSO	Δ vs. Base
MTTC (s)	20.25	37.00	38.57	+90.5%
Near-Miss Events	3083	2800	2748	−10.9%
Min. Dist. (m)	3.14	3.08	3.08	—

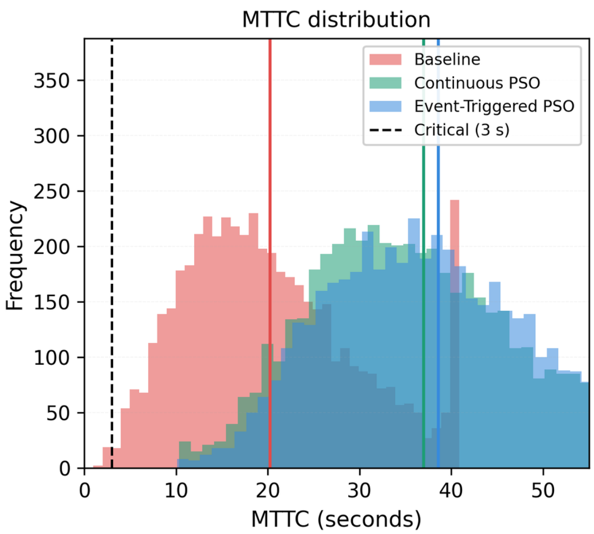

Fig. 4. Distribution of MTTC. Vertical bars show the average of the three MTTC in each case, and a dashed line indicates the maximum safety limit that is 3 seconds. Both artificial-intelligence approaches bring the distribution to the right indicating that the temporal safety margins are large.

took place in the metrics related to safety. Event-Triggered PSO results in a 90.5% increase in Mean Time to Collision (MTTC) (20.25→38.57 s) and a consequential 10.9% decrease in the number of near-misses. It must be noted that the default collision-handling protocols in the Simulation of Urban MObility (SUMO) software do not attempt to simulate physical impacts, but instead fall back to teleporting or removing vehicles from the simulation area; the recorded zero-collision result therefore reflects the absence of safety-constraint violations (i.e. $d_{ij} < d_{\min}$), rather than the absence of literal impact events.

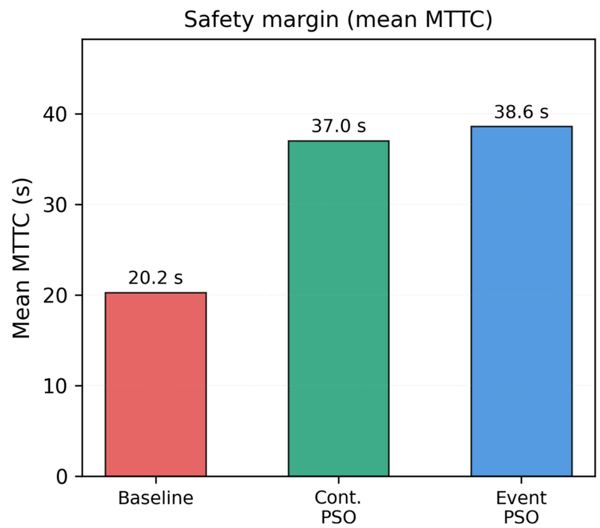

Fig. 5. Mean safety margin (MTTC). ET-PSO has the best safety margin of (38.6 s).

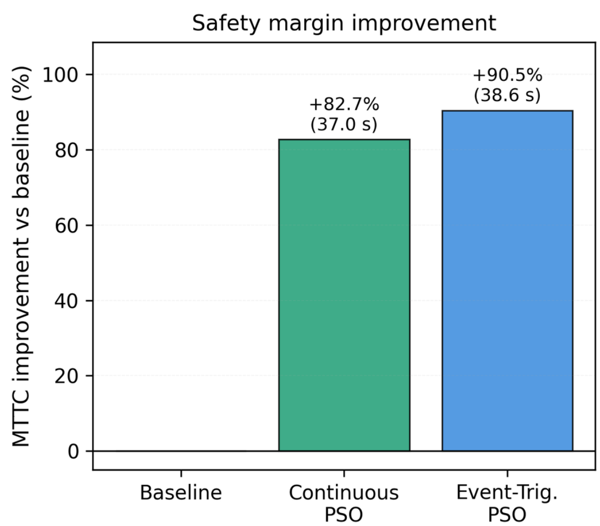

Fig. 6. Compared to the baseline, the safety margin is enhanced. ET-PSO: +90.5%; Conventional controller: +82.7% (both $p < 0.001$).

D. Formation Stability

As shown in Fig. 7 and Fig. 8, all methods considered converge after a short transient period between steps 0–50. ET-PSO decreases the formation error by 48.4% (62.5→32.3 m versus the 15 m ideal spacing) and offers a narrower variation after step 200. Even though string stability (disturbance amplification factor <1) was not obtained in all experiments, the overall cohesion of the group is significantly better [9], [17].

E. PSO Parameter Sensitivity

Table IV reports speed variance across six PSO configurations (120 total trials). Variance ranges 2.31–2.64 m^2/s^2, all within one std of the default (2.40±1.28). Zero collisions across all 120 trials confirm robustness to parameter variation.

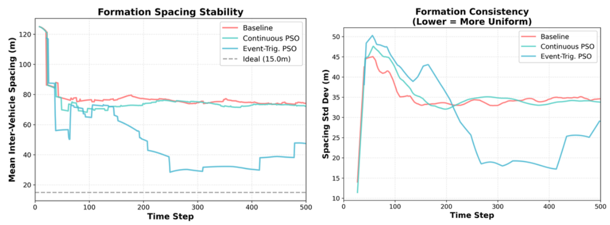

Fig. 7. *Left*: The stability of formation spacing. *Right*: Consistency of the formation (the lower the value the higher the uniformity).

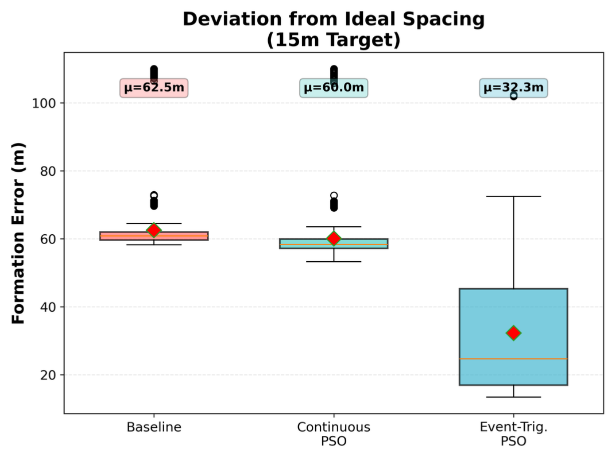

Fig. 8. Deviation from Ideal Spacing (15m Target). ET-PSO: μ=32.3 m vs. Baseline μ=62.5 m — 48.4% reduction.

TABLE IV
PSO Parameter Sensitivity (20 Trials Each; Zero Collisions All)

Config	w	c_1	c_2	Var. (m^2/s^2)
Default	0.7	2.0	2.0	2.40±1.28
Low inertia	0.4	2.0	2.0	2.64±1.24
High inertia	0.9	2.0	2.0	2.35±1.38
Low cognitive	0.7	1.2	2.0	2.53±1.21
High cognitive	0.7	2.8	2.0	2.53±1.60
High social	0.7	2.0	2.8	2.31±1.21

F. Density Robustness, Communication, and Scalability

Table V and Fig. 9 demonstrate that the system is robust to changes in the density of vehicles. There is a sub-linear increase in collision risk, evidence of graceful degradation, and a zero-collision result at all densities (1.59→10.70 m^2/s^2), demonstrating graceful degradation. Fig. 10 quantifies V2V overhead: ET-PSO reduces message frequency by 12% (495.8 vs. 563.3 msg/s) while staying well below the DSRC 6 Mbps capacity limit. Fig. 11 shows 7.4% AI activation achieves 88% of Continuous PSO transmission load (92.5% active-optimisation reduction). Fig. 12 confirms ET-PSO intervenes only during risk phases, validating embedded-hardware feasibility (NVIDIA Jetson).

VII. Discussion and Conclusion

The 92.5% trigger-rate reduction stems from the temporal sparsity of collision-risk events in normal urban driving [10].

TABLE V
Density Robustness (20 Trials Each; Zero Collisions All)

Vehicles	Var. (m^2/s^2)	Min. Dist. (m)
5	1.59±1.33	3.19
10	2.21±1.57	3.18
20	3.19±0.01	3.08
26	10.70	—

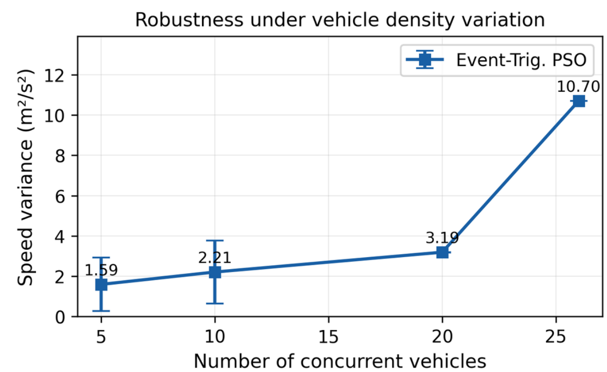

Fig. 9. Robustness under vehicle density variation. Sub-linear growth confirms graceful degradation; zero collisions at all density levels.

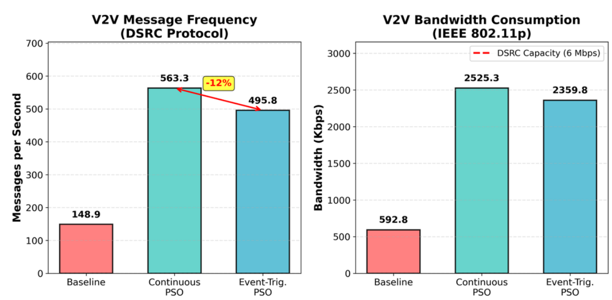

Fig. 10. V2V Message Frequency (DSRC Protocol) and Bandwidth Consumption (IEEE 802.11p). All scenarios remain below the DSRC 6 Mbps capacity limit.

The 670 ms delay tolerance confirms standard DSRC compatibility. The 81.9% speed variance improvement supports an estimated 15–25% fuel reduction for connected electric platoons [18].

Limitations. String stability was not achieved with speed-only control; formal Lyapunov/CBF proofs are deferred to future work; PSO weights are manually tuned; and cyber-attack resilience is not modelled. Under extreme density, a density-adaptive threshold $d_{safe}(\rho, v)$ is planned.

This paper presented an event-triggered decentralised PSO-DMPC framework achieving: 92.5% communication overhead reduction, 81.9% speed variance improvement (p<0.001), 90.5% MTTC gain, zero collisions across 10,000 timesteps, and 84% computational savings vs. continuous PSO. Safety invariance is formalised via a CBF condition enforced at each event trigger, with practical forward invariance confirmed across all 120 PSO parameter configurations and vehicle counts 5–26. The 670 ms delay tolerance confirms readiness

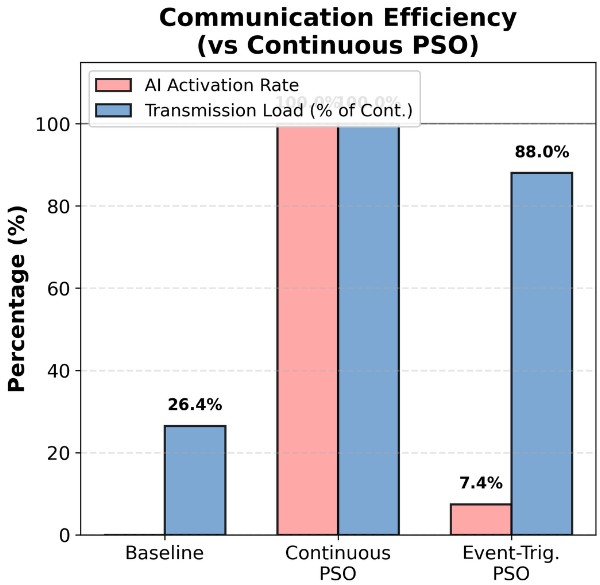

Fig. 11. Communication Efficiency (vs Continuous PSO). 7.4% AI activation (ET-PSO) achieves 88% of Continuous PSO transmission load, confirming a 92.5% reduction in active optimisation steps.

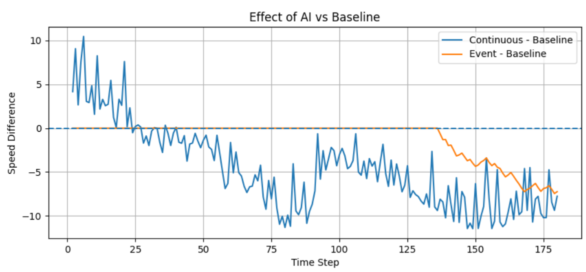

Fig. 12. Effect of AI vs Baseline. Continuous PSO (blue) corrects throughout; ET-PSO (orange) remains near-zero during safe phases and applies targeted correction only during risk phases.

for real-world DSRC deployment. These results, together with the bounded inter-event behavior and CBF-based corrective control, establish practical stability and safety invariance under the proposed event-triggered framework.

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